CS767 Assignment 2 — Data Analysis Agent

Student: Dongqing Yang

ID: *291140653*

GitHub: *https://github.com/yangdongqing214/compsci767_assignment2*

Demo video: *link in README*

Page 1 — System design

Repository

https://github.com/yangdongqing214/compsci767_assignment2

Architecture

The agent follows a **perceive → decide → act → reflect** loop:

1. **Perceive:** Load CSV; compute profile (columns, types, missing counts).
2. **Decide:** Planner produces a JSON list of steps (LLM if `OPENAI_API_KEY` set, else heuristic rules).
3. **Act:** Each step runs a **builtin tool** (statistics, plots, groupby) or **sandboxed pandas code** validated by `safety.py`.
4. **Reflect:** Results stored in `AgentMemory`; final markdown report written to `output/`.

```
User goal + CSV
      ↓
 [DataAnalysisAgent]
      ↓
 profile → plan → execute steps → memory + report.md + PNG artifacts
```

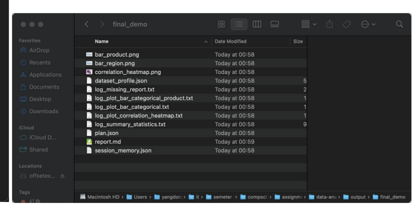
Design choices

Tools vs. free-form code: Most steps use reliable builtins; optional code step for goal-specific metrics (e.g. revenue = units × price).

Safety: AST scan blocks os, subprocess, open, network imports before exec.

Reproducibility: Heuristic mode requires no API; graders can run python main.py immediately.

Screenshot 1,2 — CLI run, Generated artifacts



1.Completes 5—6 steps and prints a short summary.

2.Histogram, correlation heatmap, and bar charts produced automatically.

How it works (short)

The user supplies a **goal** and **CSV**. The agent does not hard-code one analysis pipeline—it **selects steps from the dataset profile** and user goal, executes them, and aggregates results. Failed steps are logged without stopping the whole run (robust act loop). **For limitations**, single-pass planning — no observe→replan if a step fails. Memory logs outcomes but does not feed back into replanning within one run. Trade-off: builtin tools improve reliability over fully LLM-generated code.